

**A binary search
to find the letter “P”
in the letter table**

LetterTable

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26

$$\begin{array}{c} \text{Lower} \\ \boxed{0} \end{array} \quad \begin{array}{c} \text{Upper} \\ \boxed{0} \end{array} \quad \begin{array}{c} \text{Middle} \\ \boxed{^e 0} \end{array} = \begin{array}{c} \text{Letter(Middle)} \\ \boxed{} \end{array}$$

MOVE 1 TO Lower

MOVE 26 TO Upper

PERFORM UNTIL ItemFound OR ItemNotInTable

 COMPUTE Middle = (Lower + Upper) / 2

 EVALUATE TRUE

 WHEN Letter(Middle) < "P" COMPUTE Lower = Middle + 1

 WHEN Letter(Middle) > "P" COMPUTE Upper = Middle - 1

 WHEN Letter(Middle) = "P" SET ItemFound TO TRUE

 WHEN Lower > Upper THEN SET ItemNotInTable TO TRUE

 END-EVALUATE

END-PERFORM

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Lower	Upper	Middle		Letter(Middle)
1	26	0	=	

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